# Caleb Kilgo

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## Education

**University of Alabama in Huntsville** — Huntsville, AL

*Bachelor of Science in Computer Science, Minor in Mathematics, Focus in Data Science* — *Aug. 2025 – May 2029*

- **GPA: 3.75 / 4.00**
- Relevant Coursework: Computer Science II (Data Structures), Computer Science I (C++), Discrete Structures, Introduction to CS using Python, Calculus I, Calculus II

## Experience

**Undergraduate Data Science Research Assistant** — Sep. 2025 – Present

*University of Alabama in Huntsville* — *Huntsville, AL*

- Researched automated requirement-to-code traceability for safety-critical NASA and DoD flight software, targeting a manual verification bottleneck in software assurance.
- Built an aerospace traceability dataset from NASA's Core Flight System (cFS) Checksum app: 284 ground-truth links across 114 requirements and 65 functions.
- Fine-tuned the CoCoSoDa contrastive transformer with InfoNCE and BM25-mined hard negatives, achieving 0.58 Test MRR and 0.74 Recall@10 under leave-one-group-out CV.

**Information Technology Assistant** — May 2024 – Aug. 2024

*Fort Payne City Schools* — *Fort Payne, AL*

- Performed diagnostics, repairs, and maintenance on over 500 Chromebooks, resolving hardware failures and restoring device functionality.
- Conducted inventory management and asset tracking of all Chromebooks across the Fort Payne City School System using Google Sheets.
- Responded promptly to IT support tickets, delivering efficient technical assistance to resolve software, network, and peripheral device issues.

## Projects

**Real-Time Satellite Tracker** | *React, Vite, C++, WebAssembly, FastAPI* — Jan. 2026 – Present

- Built a browser-based 3D platform rendering thousands of satellites in real time from live TLE data via a batched CesiumJS PointPrimitiveCollection.
- Accelerated SGP4 propagation by compiling a custom C++ engine to WebAssembly, sustaining smooth frame rates across the full catalog.
- Deployed on Vercel with push-to-deploy and an interactive interface for filtering, selection, live telemetry, and camera lock-on.

**Automobile Multimodal RAG Assistant** | *Python, Ollama, SigLip2, Streamlit* — Jan. 2026 – Present

- Developed a local-first multimodal RAG system for automotive manuals, dynamically ingesting and indexing variable-length manuals with joint text and image FAISS indices for grounded retrieval.
- Integrated SigLip2-based image embeddings and a vision-language model (VLM) to surface all relevant diagrams and figures per query, enabling step-by-step procedural guidance directly grounded in manual content.
- Built an offline-capable end-to-end pipeline (PDF parsing, chunking, embedding, retrieval, and generation) with low-latency inference and modular design to support additional manuals, models, and UI extensions.

## Technical Skills

**Languages**: Python, C++, JavaScript, HTML/CSS
**Frameworks**: React, Vite, FastAPI, CesiumJS/Resium, WebAssembly
**Developer Tools**: Git, Docker, Vercel, Google Cloud Platform, VS Code, Visual Studio, PyCharm, IntelliJ, Jupyter
**Libraries**: PyTorch, NumPy, Pandas, Matplotlib, Plotly

## Clubs and Organizations

**UAH i4 Program** | *Member* — June 2026 – Present

**Data Science and Artificial Intelligence Club (DSAI)** | *Member* — Aug. 2025 – Present

**Association of Computing Machinery (ACM)** | *Member* — Aug. 2025 – Present